Validation Report

1 Job data

Type	Location	Reference
Validation report	Leiria	MEAU-DataAcqBoard

Done by	Start date	End date
Joao Gaspar	24/02/2024	24/02/2024

Brief
The current document provides a summary of the validation tests/checks done in the context of MEAU-DataAcqBoard project. Each solution is evaluated OK/NOK depending on whether the validation results met/didn't meet the established requirements.

2	Requirements
2.1	Develop microcontroller based module (familiarity, ease of deployment, enough processing power)
Solution **OK**	Selected **ESP32** as processing unit for the module. Robust hardware and firmware design support.

2.2	Capable of acquiring at least 10 analog signals (NTCs) and 1 digital signal (flow meter)
Solution **OK**	Resorted to **both ESP32 GPIOs and ADC expander** to acquire a total of 13 inputs. • 10 NTC sensor inputs (analog) • 1 Flow Meter input (digital) • 1 Generic input (analog/digital) • 1 User START button (digital)

2.3 Analog signal voltage reading error must not exceed 5%

Solution

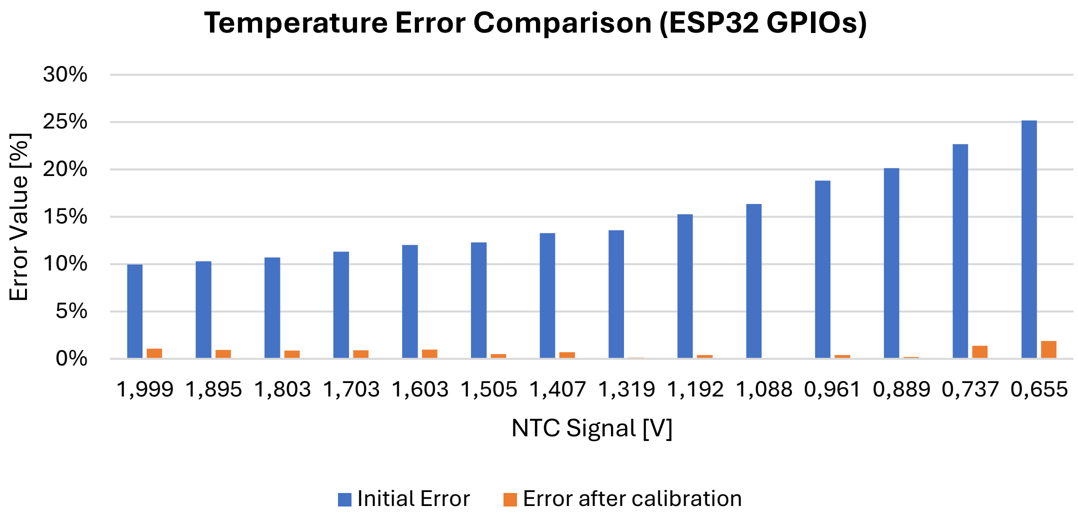
OK

Calibration tests were run to determine voltage offset between the values obtained by the module and the real NTC voltage value measured with oscilloscope.

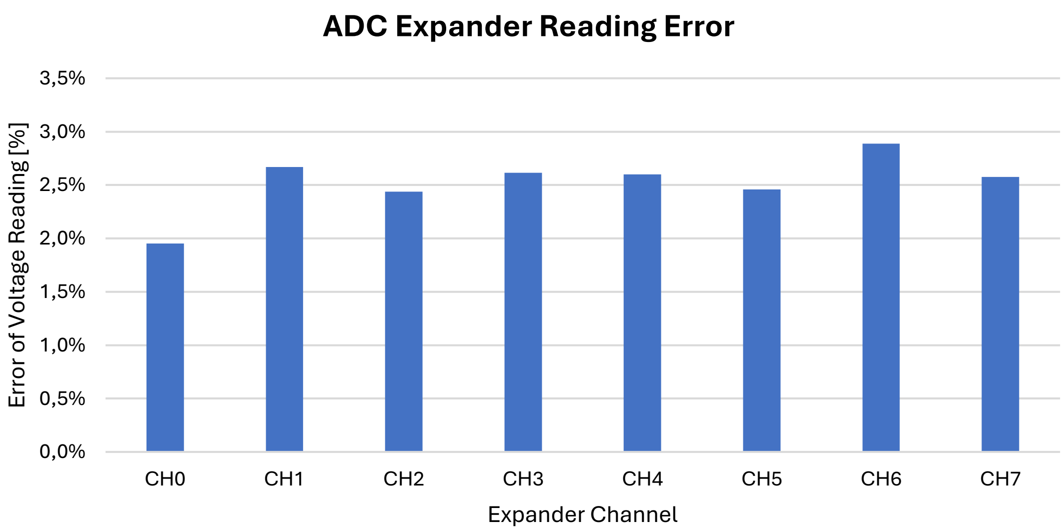
Calibration tests consisted in:

1. Submerging the NTC that was connected to the module in boiling water.
2. Voltage values from the ESP32 GPIOs, the ADC expander and the oscilloscope were registered at **intervals of 0.1V** of NTC signal voltage.
3. Only voltages between **2V and 0.65V** were considered, since this signal voltage range represented NTC temperature readings ranging from **15ºC to 65ºC**, considering the module's hardware configuration.

Temperature error percentage before and after correction factors were introduced are compiled below:



The validation tests revealed great accuracy from the ADC expander, having been measured an average error across the relevant voltage range and across all channels of 2,52%. Below there follows a chart of voltage reading error per expander channel:



2.4	Capable of storing the acquired data in permanent storage at 2Hz
Solution **OK**	Verified with data from log files. Time delta between each stored sample matches the requirement.

xtime[s]	pressure_delta[Pa]	flow_rate[kg/s]	flow_rate[l/min]	inlet_temp[C]	oulet_temp[C]	ntc0_temp[C]	ntc1_temp[C]	ntc2_temp[C]
0	1,18	0,0265	1,693	25,093	25,254	25,661	24,809	25,497
0,5	1,19	0,02686	1,716	25,056	25,168	25,707	24,867	25,544
1	1,19	0,02686	1,716	25,075	25,18	25,591	24,847	25,591
1,5	1,17	0,02771	1,77	25,087	25,285	25,591	24,847	25,591
2	1,17	0,02771	1,77	25,1	25,162	25,614	24,828	25,615
2,5	1,17	0,02771	1,77	25,062	25,205	25,638	24,828	25,474
3	1,15	0,02778	1,775	25,093	25,149	25,591	24,886	25,638
3,5	1,17	0,02778	1,775	25,106	25,236	25,521	24,847	25,615
4	1,21	0,02778	1,775	25,075	25,248	25,638	24,828	25,591
4,5	1,17	0,02718	1,736	25,1	25,205	25,707	24,867	25,591
5	1,18	0,02718	1,736	25,081	25,168	25,521	24,828	25,474
5,5	1,17	0,02718	1,736	25,106	25,254	25,638	24,867	25,544
6	1,13	0,02716	1,735	25,087	25,279	25,568	24,789	25,544
6,5	1,19	0,02716	1,735	25,093	25,285	25,591	24,789	25,521
7	1,17	0,02716	1,735	25,081	25,316	25,684	24,828	25,427

2.5	Capable of displaying acquired data to user in a timely manner (test safety control in real time)
Solution **OK**	Verified in the timestamps of each UART message. Time delta between each message stayed at a steady 500ms during a continuous 20 minute test.

2.6	Hardware to enable manual control of data acquisition start/stop
Solution **OK**	Deployed a user button which was set as a toggle for the current data acquisition state in firmware.

3	Optional Features
3.1	Implementation of CAN Bus to allow future use as datalogger of data generated elsewhere
Solution OK	CAN support hardware (transceiver and noise filter) were deployed in board design. Validated with a basic CAN example. Not in use in the firmware version that will run during the BTMS bench test.

3.2	Power outputs (+12V) to allow board to drive cooling accessories. Water Pump output is to be used to generate a PWM signal to set pump speed only. (Output current rating: 1A) Fan output is to be used to supply power to a fan with a current rating of 0.3A. (Output current rating: 1A) Extra power output may be used to power other loads. (Output current rating: 2A)
Solution OK	Implemented a total of 3 high-side MOSFETs actuated with MOSFET drivers for greater reliability. All power outputs were submitted to a basic load test at their current ratings. Load tests consisted in: 1. Supplying +12V to a power resistor for 15 minutes with each individual power output. 2. A power resistor of 10Ω was used to test the Water Pump and Fan outputs. 3. A power resistor of 5Ω was used to test the Extra output. Having tested each individual output board temperature rise did not exceed 10ºC. Not in use in the firmware version that will run during the BTMS bench test.